

MIS 3353 – Databases

Section 003 | Fall 2026

Course Information

Instructor:	Trevon Holt
Office:	Adams Hall 3101
Office Hours:	Tuesday 10:30 AM–1:00 PM; Thursday 10:30 AM–1:00 PM (Open Door Policy)
Location:	Adams Hall 3106
Meeting Times:	Tue/Thu 9:00 AM–10:15 AM
Course Site:	https://canvas.ou.edu
Final Exam:	Mon, Dec 14, 2026, 8:00 AM–10:00 AM, Adams Hall 3106

Prerequisites

MIS 2113 – Computer-Based Info Systems

Textbooks & Software

[Required] *Textbook:* Essentials Plus of SQL: Extracting and Updating Data — Available Via Canvas or the OU Bookstore

[Required] *Textbook:* TopHat — Details Provided in Canvas

[Required] *Software:* LucidChart — Full access to the software is available for free when you log in with your OU email.

[Required] *Software:* SQL Server using Microsoft SQL Server Management Studio. — More information will be given in class.

Course Description

MIS 3353 Database Management provides a strong foundation for understanding organizational information use and management. Concepts discussed are vital to anyone pursuing a career in information systems management, data analysis, or any field where understanding and use of stored data is important (e.g., most business fields).

We will cover important concepts and skills useful in the rest of your coursework as well as your

future careers, including:

The fundamentals of database use and design:

SQL to interact with functional databases.

Conceptual, logical, and physical design of databases.

Database implementation.

Organizational issues involved in data management.

Basic business cycles (revenue, expenditure, production) and the data and database structures needed to support them.

This class is not easy. You should expect to expend considerable time and effort. However, it can be fun learning the different aspects of data management. Stay up-to-date on the course material and do not hesitate to ask the instructor any questions you may have about the material as the course progresses.

Course Design

This class combines a variety of techniques to capture student interests from several possible angles. The readings and lectures provide a base of knowledge that students can use in other components of the course. In-class exercises provide an avenue for applying the knowledge base. Class discussions provide an opportunity for discovery. This approach only works when students complete the readings before class and then participate in the exercises and discussions.

Course Level Outcomes (CLOs)

CLO-1: Extract and Analyze Data Using SQL: Retrieve and manipulate data from relational databases using SQL queries that incorporate joins, subqueries, built-in functions, and aggregation. Apply data aggregation and filtering techniques to summarize datasets for decision-making. *[Assessed by: Participations 1-7, Quiz Chapters 1-4, Homework 1-2, Exam I]*

CLO-2: Design and Model Relational Databases: Develop and interpret Entity-Relationship Diagrams (ERDs) to model business processes, including one-to-many, many-to-many, and hierarchical relationships. Translate conceptual models into logical and physical database designs. *[Assessed by: Participations 8-13, Quiz Chapters 5-11, Homework 3-4, Exam II]*

CLO-3: Apply Normalization and Implement Databases: Explain and apply normalization principles up to Third Normal Form (3NF) to eliminate redundancy and improve database performance. Design and implement normalized database schemas and execute CRUD operations to populate

and manage data in SQL Server. *[Assessed by: Participations 14-18, Quiz Chapter 8, Homework 5, Final Exam]*

CLO-4: Compare and Utilize Advanced Data Management Systems: Compare SQL-based systems with modern alternatives such as NoSQL, Data Warehouses, Data Lakes, and Blockchain in terms of structure, scalability, and use cases. Use MongoDB to query and manage JSON data and demonstrate blockchain data concepts through applied exercises. *[Assessed by: Participations 19-20, Homework 6, Final Exam]*

CLO-5: Integrate Database Knowledge for Business Decision-Making: Combine database extraction, design, and implementation skills to solve realistic business data problems. Evaluate data management solutions for scalability, integrity, and efficiency in organizational contexts. *[Assessed by: Participations (11, 12, 17, 18), Exam I, Exam II, Final Exam]*

Grading & Assessments

Assessment Breakdown

Class Participation and Roll-Call Attendance – 5% (2 lowest assignment scores will be dropped)

Quizzes – 5% (2 lowest quiz scores will be dropped)

Homework – 20%

Module Reflections – 10%

Exams – 60%

Class Participation and Roll-Call Attendance

For the design of this class to work, students must be actively involved in the class. Active involvement consists of three components: (1) class attendance; (2) reading and thinking about the material before coming to class; and (3) sharing of and discussing thoughts and ideas. Students' participation grades are dependent on all of these, culminating on the third item.

To receive participation (2.5%), you must do the participation exercises scheduled for each class.

To receive attendance (2.5%), you must be present in class

Please realize that, during class discussions and exercises, it is okay to be wrong! You will get a participation point whether you are correct or not. If you try to answer a question, and your answer is incorrect, there is a good chance you are not the only one who thought that way. By seeing the wrong answers and understanding why they are wrong, everyone can learn. Wrong answers on

exams, of course, still carry a penalty, so it's important to get those answers sorted out during class time.

Quizzes

To provide an incentive for students to come to class prepared, quizzes are assigned that correspond with reading assignments. These quizzes will be available via Canvas and are due before class on the date noted in the calendar at the end of this syllabus. These quizzes are open-book and must be taken individually, with no help from other students or individuals. Each quiz may be taken twice. The two lowest quiz scores will be dropped in evaluating each student's semester quiz grade.

Homework

Homework assignments for this course are designed to encourage students to work actively with the course material and thereby master course materials. Further, homework enables students and the instructor to recognize any points that are not yet fully understood. Students are expected to complete each of the homework problems. If, after considerable effort, you are stuck on a problem, note (on the homework) the specific difficulty you are having and move on.

Students may discuss homework problems and approaches to answering them with their classmates. However, students are expected to work independently in the actual completion of their work. For instance, if two (or more) homework answers have the same typos, it will be assumed that work was not done independently and therefore represents an honor code violation. In such cases, the instructor will take appropriate action, ranging from a reduction in grade to reporting of the violation. More importantly, it is essential that you understand the homework assignments in order to do well on the exams.

Module Reflections

The Reflection should summarize the student's learning experience (engagement with materials, quizzes, participation, etc.) in two paragraphs with a minimum of 10 lines per paragraph. There are 3 parts to the reflection:

1. Summarize the module's learning materials (readings, media, recorded lectures, and Live Session) and how they relate to the module learning objectives.
2. List two topics of interest from the module's content using the following questions as guidance:
3. What are topics that are most interesting to you?
4. What are topics that can be utilized immediately at your work?
5. What is a question that you still have about this module's learning topics?

Exams

There will be two midterm exams and one comprehensive final exam. All exams will consist of multiple-choice, short answer problems, writing SQL queries, etc. These exams are designed to assess your basic understanding of the concepts and their application. These exams are scheduled in-person. Please check Canvas for any updates. Students are required to bring their laptop to the exams.

Grading Scale

A: 90–100%

B: 80–89%

C: 70–79%

D: 60–69%

F: 0–59%

Course Policies

Timeliness / Late Work

To receive full credit for a written assignment, assignments must be turned in before the beginning of the class period on the day it is due. **Late work will not be accepted.** If you have obligations that conflict with exam or assignment due dates, you should make arrangements with the instructor as soon as possible to complete these ahead of time. Feel free to submit homework assignments early.

Grade / Score Appeals

Students wishing to appeal a grade or score **must submit the appeal to the instructor in writing within 48 hours of receiving the graded work.** Submission of appeals must be sent via Microsoft Teams/Email. Score changes are at the discretion of the instructor, not the TA. It is important to understand that I may require an in-person meeting to continue the conversation.

If you would like to talk about your overall grade in the course, you must find a time during office hours or schedule an appointment to meet **in-person**. **I will NOT respond to emails asking for a grade bump.**

Missed Exams / Make-Up Policy

Make-up exams are only available under extraordinary circumstances (e.g., documented medical

emergency, family emergency). If you need to miss an exam, you must contact the instructor **at least 48 hours before** the scheduled exam and provide documentation. Conflicts known in advance (e.g., travel, other exams) do not qualify as extraordinary circumstances.

Incomplete Grades

A grade of Incomplete (I) may be awarded only when a student has been unable to complete coursework due to a legitimate emergency that occurs near the end of the semester.

Incompletes will not be awarded to students who simply need more time to master the material or to improve their grade. Any request for an incomplete must be discussed with the instructor prior to the end of the semester.

Communication

Please call me Professor or Professor Holt. When you send an e-mail to me, please be sure to include the course number or some course identifier in the Subject line. Additionally, please identify yourself in the text of your e-mail, otherwise, I may not figure out who you are just by your email address alone. Please note that if your question is relevant to the class, I will forward your email with my response to the class – so be sure to proofread your email and use good email etiquette because it may be seen by your classmates. It is the student's responsibility to receive/check incoming mail in a timely fashion (e.g., not regularly checking your email, having a “full mailbox” condition, etc.).

You can expect a response to emails sent Monday-Friday within 24 hours. Emails sent over the weekend will receive a response within 48 hours. We will use Microsoft Teams for course communication. Please send a message through Teams direct messaging.

NOTE: When a student has a question or needs to inform the instructor of an important event, you must contact me via Microsoft Teams/Email.

University Policies

Mental Health Support Services

As a university student, you may experience a range of issues that can cause barriers to learning, such as strained relationships, increased anxiety, alcohol/drug problems, feeling down, difficulty

concentrating and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. OU offers services to assist you with addressing these and other concerns you may be experiencing. You can learn more about the broad range of confidential mental health services available on campus via the Counseling, Wellness, and Prevention Services (CWPS) at <https://www.ou.edu/health/cwps>, (405) 325-2911 or at Goddard Health Center, 620 Elm Avenue. Additionally, you can also access TimelyCare, a virtual health and well-being platform that provides 24/7/365 access to medical care, mental health support, and wellness services. To connect with TimelyCare, visit [timelycare.com/OU](<https://timelycare.com/OU>) or download the app.

Title IX / Sexual Misconduct

The University of Oklahoma is committed to providing a safe learning environment for all members of our community. Title IX of the Education Amendments of 1972 prohibits sex discrimination in educational programs and activities. This includes sexual harassment, sexual assault, domestic violence, dating violence, and stalking.

As your instructor, I am a **mandatory reporter**. This means that if you share information about sexual misconduct or harassment with me – whether in person, in writing, in class discussions, on Canvas, or in any other setting – I am required to report that information to the OU Institutional Equity Office. If you need confidential support, you may contact **OU Advocates** at (405) 615-0013. You can also reach the Institutional Equity Office at (405) 325-3546 or visit <https://www.ou.edu/eoo>.

Reasonable Accommodation

The University of Oklahoma is committed to making its activities accessible to all participants. If you have a disability and require accommodations in this course, please contact the **Accessibility and Disability Resource Center (ADRC)** through iAdvise to schedule an appointment at <https://www.ou.edu/adrc>, by emailing adrc@ou.edu, or by calling (405) 325-3852. Accommodations are not retroactive. You must provide your accommodation letter to me as early as possible in the semester to ensure appropriate arrangements can be made.

Religious Observance

It is the policy of the University to excuse the absences of students that result from religious observances and to provide without penalty for the rescheduling of examinations and additional required classwork that may fall on religious holidays [Faculty Handbook 3.15.2]. If a religious holiday conflicts with a class meeting, quiz, exam, or assignment deadline, please notify me at least one week in advance so that we can make appropriate arrangements.

Pregnancy Adjustments

Title IX protects students from discrimination based on pregnancy, childbirth, or related conditions. Students who are pregnant or experiencing pregnancy-related conditions may request reasonable adjustments. For assistance, contact the **Institutional Equity Office** at (405) 325-3546 or visit <https://www.ou.edu/eoo>.

Final Exam Preparation Period

The 7 calendar days before the first day of the final exam period are designated as the final exam preparation period. During this time, no major homework, projects, presentations, or other graded work may be assigned that is due before the final exam period begins. Instructors may continue to cover new material during this period. Please refer to the Academic Calendar for the specific dates each semester.

Emergency Protocol — Severe Weather

In the event of severe weather, follow OU Alert instructions. If a tornado warning is issued while you are on campus:

- **Get In** — Move to the nearest interior shelter location.
- **Get Down** — Go to the lowest floor possible.
- **Cover Up** — Protect your head and neck.

Building-specific severe weather refuge maps are posted throughout campus. Sign up for **OU Alert** at <https://www.ou.edu/oupd/oualert> to receive emergency notifications.

Emergency Protocol – Active Threat (Run, Hide, Fight)

In the event of an active threat on campus:

- **Run** – If there is a safe escape path, leave immediately and call 911.
- **Hide** – If evacuation is not possible, find a secure location, lock/barricade the door, and silence your phone.
- **Fight** – As a last resort, use whatever means necessary to disrupt and disarm the threat.

OU Alerts are sent via the one.ou.edu system. Emergency contacts: OUPD Norman (405) 325-1717 / (405) 325-1911; OUPD Tulsa (918) 660-3900 / (918) 660-3333.

Emergency Protocol – Fire Alarm

In the event a fire alarm sounds:

1. **Leave** the building immediately.
2. **Know** your nearest exit route.
3. **Assist** those who need help evacuating, if it is safe to do so.
4. **Proceed** to the designated assembly area.
5. **Notify** emergency personnel of anyone who could not evacuate.
6. **Wait** for official clearance before re-entering the building.

Copyright Statement

Course materials prepared by the instructor are the intellectual property of the instructor. Students are expressly prohibited from selling notes or other course materials to any person or commercial firm without the express written permission of the instructor. Recording of lectures or class sessions is permitted only with the instructor's permission. Any approved recordings are for personal academic use only and may not be distributed, published, or posted publicly in any form.

Generative AI Policy

I permit the use of standalone Generative AI tools like ChatGPT, Copilot, Claude, etc., for the

following:

- Searching for publications on your research topic.
- Creating an outline, text, or speech for presentations and writeups.
- Providing grammatical feedback on drafts.
- Further study on concepts outside of class.
- Troubleshooting problems with software.
- Creating initial ideas for project.

Whenever you use Generative AI for any of these, you are required to make a declaration acknowledging how and to what extent you used it. Please include an appendix that details the prompts you used and the responses you obtained. Usage of Generative AI outside of the clearly defined scope, and without acknowledgment as required, will be considered a violation of the academic integrity policy for this course.

Course Calendar

The schedule below is tentative; adjustments may be made. Announcements in class, via email, or on Canvas supersede this outline.

Wk	Tue	Thu
1	<u>Tue, Aug 25, 2026</u> Course Overview & Intro to Database <i>Syllabus & Chapter 1, MyEducator & Top Hat</i> Quiz 1 Syllabus DUE -Read Chapter 1 Top Hat	<u>Thu, Aug 27, 2026</u> Extracting Data: The SELECT Statement <i>Chapter 2, MyEducator & Top Hat</i> Quiz Chapter 2 TopHat
2	<u>Tue, Sep 1, 2026</u> Extracting Data: The SELECT Statement <i>Chapter 2, MyEducator & Top Hat</i> Quiz Chapter 3 Top Hat PARTICIPATION 1 – SIMPLE SQL	<u>Thu, Sep 3, 2026</u> Data Across Tables: The JOIN Statement <i>Chapter 3, MyEducator & Top Hat</i> PARTICIPATION 2 – INNER JOIN
3	<u>Tue, Sep 8, 2026</u> Compounding Data & Start JOINS <i>Chapter 3 Top Hat Quiz Chapter 4 Top Hat</i> PARTICIPATION 3 – DATA AGGREGATION & JOIN	<u>Thu, Sep 10, 2026</u> Different Types of JOIN <i>Chapter 5 MyEducator & Chapter 4 Top Hat</i> HW#1 – SQL due PARTICIPATION 4 – DIFFERENT TYPES OF JOINS
4	<u>Tue, Sep 15, 2026</u> Built-in Functions & Subqueries <i>MyEducator & Chapter 4 Top Hat</i> PARTICIPATION 5 – BUILT-IN FUNCTIONS PARTICIPATION 6 – SUBQUERIES	<u>Thu, Sep 17, 2026</u> Subqueries & Review <i>MyEducator & Chapter 4 Top Ha</i> PARTICIPATION 6 – SUBQUERIES (cont.) HW#2 – SQL due PARTICIPATION 7 – SQL REVIEW

Wk	Tue	Thu
5	<u>Tue, Sep 22, 2026</u> Exam-1 SQL	<u>Thu, Sep 24, 2026</u> Intro to Data Modeling & ERDs <i>Chapter 5 Top Hat Quiz Chapter 5 Top Hat</i> REFLECTION 1 DUE PARTICIPATION 8 – INTRO TO ERDS
6	<u>Tue, Sep 29, 2026</u> More ERDs PARTICIPATION 9 – INTERMEDIATE ERDS	<u>Thu, Oct 1, 2026</u> Enhanced ERDs <i>Chapter 6 Top Hat</i> PARTICIPATION 10 – EERD Quiz Chapter 6 Top Hat HW#3 – ERD due (use Lucidchart to submit pdf on Canvas)
7	<u>Tue, Oct 6, 2026</u> Revenue Cycle <i>Chapter 7 Top Hat</i> PARTICIPATION 11 – REVENUE CYCLE Quiz Chapter 7 Top Hat	<u>Thu, Oct 8, 2026</u> Expenditure Cycle <i>Chapter 7 Top Hat</i> PARTICIPATION 12 – EXPENDITURE CYCLE
8	<u>Tue, Oct 13, 2026</u> Production Cycles & More ERDs <i>Chapter 7 Top Hat</i> PARTICIPATION 13 – MORE ERDS	<u>Thu, Oct 15, 2026</u> Review HW#4 – Advanced ERD due (use Lucidchart submit pdf on Canvas)
9	<u>Tue, Oct 20, 2026</u> EXAM- 2 ERD	<u>Thu, Oct 22, 2026</u> Logical Design and Normalization <i>Chapter 8 Top Hat</i> Quiz Chapter 8 Top Hat REFLECTION II DUE PARTICIPATION 14 – LOGICAL DESIGN
10	<u>Tue, Oct 27, 2026</u> Logical Design <i>Chapter 8 Top Hat</i> PARTICIPATION 14 – LOGICAL DESIGN	<u>Thu, Oct 29, 2026</u> Normalization <i>Chapter 8 Top Hat</i> PARTICIPATION 15 – NORMALIZATION
11	<u>Tue, Nov 3, 2026</u> Physical Design <i>Chapter 9 Top Hat</i> Quiz Chapter 9 Top Hat HW#5 – Normalization due (submit pdf on Canvas)	<u>Thu, Nov 5, 2026</u> Implementation <i>Chapter 10 Top Hat</i> PARTICIPATION 16 – IMPLEMENTATION Quiz Chapter 10 Top Hat
12	<u>Tue, Nov 10, 2026</u> Implementation <i>Chapter 10 Top Hat</i> PARTICIPATION 17 – DATA CLEANING & IMPLEMENTATION	<u>Thu, Nov 12, 2026</u> <i>Chapter 10 Top Hat</i> Implementation due (implement on SQL Server & submit code on Canvas) PARTICIPATION 18 – DATA CLEANING & IMPLEMENTATION
13	<u>Tue, Nov 17, 2026</u> Data Domiciles <i>Chapter 10 Top Hat</i> Quiz Chapter 11 Top Hat REFLECTION III due	<u>Thu, Nov 19, 2026</u> Introduction to Data Warehousing & NoSQL SET UP MONGODB – at home Querying MongoDB
14	<u>Tue, Nov 24, 2026</u> Querying MongoDB	<u>Thu, Nov 26, 2026</u> Thanksgiving Break

Wk	Tue	Thu
	PARTICIPATION 19 – QUERYING MONGODB	<i>No Class</i>
15	<u>Tue, Dec 1, 2026</u> HW#6 – Querying MongoDB (run on MongoDB & submit code on Canvas)	<u>Thu, Dec 3, 2026</u> Blockchain Game PARTICIPATION 20 – BLOCKCHAIN GAME
16	<u>Tue, Dec 8, 2026</u> Flow over sessions	<u>Thu, Dec 10, 2026</u> Review and Practice for Final REFLECTION IV due
Final	FINAL EXAM Mon, Dec 14, 2026 · 8:00 AM – 10:00 AM · Room: Adams Hall 3106	